

Decision and Risk Analysis

Decision Rule Problems

DAEN 427 / ISEN 427 / ISEN 627

Alexander Abuabara

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Formulas

Probability model

The prior and data produce the probability to the decision:

$$q = P(H = 1 \mid D).$$

For a next case, q may be posterior predictive.

For a process claim, q may be a posterior tail probability.

Loss model

With zero loss for correct decisions,

$$\mathcal{R}(a_+) = C_{\text{FP}}(1 - q),$$

$$\mathcal{R}(a_-) = C_{\text{FN}}q.$$

Take the positive action when

$$q > \tau, \quad \tau = \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}},$$

$$\tau = \frac{1}{1 + r},$$

where $r = C_{\text{FN}}/C_{\text{FP}}$.

Formulas

Question 1: next case

Decision target:

$$Y_{\text{new}} = 1.$$

Probability:

$$P(Y_{\text{new}} = 1 \mid D) = E(\theta \mid D).$$

Question 2: process state

Decision target:

$$\theta > \theta_0.$$

Probability:

$$P(\theta > \theta_0 \mid D).$$

Q1) Screening a transaction for fraud

An online retailer models the fraud probability θ for transactions in a new risk segment. Before observing segment data,

$$\theta \sim \text{Beta}(2, 18).$$

Among the first 24 transactions, 6 were fraudulent. For the next transaction, the retailer may send it to manual review (a_+) or approve it automatically (a_-).

- a. Find the posterior distribution of θ .
- b. Find the posterior predictive probability that the next transaction is fraudulent.
- c. If $C_{FP} = \$25$ and $C_{FN} = \$180$, find the action threshold and the Bayes action. Verify the decision using expected losses.
- d. Find the critical false-negative cost above which manual review becomes optimal, holding $C_{FP} = \$25$ fixed.

Q1a ... posterior distribution

Step 1. Identify the prior parameters and observed counts:

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Step 3. Substitute the values:

$$\theta \mid D \sim \text{Beta}(2 + 6, 18 + 18).$$

Answer:

$\theta \mid D \sim \text{Beta}(8, 36)$ The six observed frauds increase the prior fraud count from 2 to 8.

Q1b ... probability for the next transaction

The decision concerns one future binary outcome, so use the Beta–Bernoulli posterior predictive probability.

$$Y_{\text{new}} = \begin{cases} 1, & \text{if the next transaction is fraudulent,} \\ 0, & \text{if the next transaction is legitimate.} \end{cases}$$

and D represents all the observed data used to update the prior distribution.

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Step 1. Use the posterior mean:

$$q = P(Y_{\text{new}} = 1 \mid D) = E(\theta \mid D).$$

Step 2. Insert the posterior parameters:

$$q = \frac{8}{8 + 36} = \frac{8}{44} = \frac{2}{11}.$$

Answer:

$q \approx 0.1818$. The model assigns an 18.18% fraud probability to the next transaction in this segment.

Q1c ... loss table

Decision	Fraudulent ($Y_{\text{new}} = 1$)	Legitimate ($Y_{\text{new}} = 0$)
Manual review (a_+)	Correct positive: \$0	False positive: \$25
Automatic approval (a_-)	False negative: \$180	Correct negative: \$0

Q1c ... threshold and Bayes action

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Manual review has the smaller posterior expected loss.

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Step 3. Verify directly:

$$\mathcal{R}(a_+) = 25(1 - 2/11) = \$20.45,$$

$$\mathcal{R}(a_-) = 180(2/11) = \$32.73.$$

Answer: Send the next transaction to manual review.

Q1d ... critical false-negative cost

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Step 1. Solve for C_{FN} :

$$C_{FN} > C_{FP} \frac{1 - q}{q}.$$

Step 2. Substitute $C_{FP} = 25$ and $q = 2/11$:

$$C_{FN} > 25 \frac{9/11}{2/11} = 25 \left(\frac{9}{2} \right) = 112.50.$$

Answer:

Manual review is optimal when $C_{FN} > \$112.50$.

At $C_{FN} = \$112.50$, the retailer is indifferent.

The stated cost of \$180 lies above the boundary.

Q2) Intervening in a production process

A food plant tracks the contamination rate θ for a production line. The prior is

$$\theta \sim \text{Beta}(2, 8).$$

Testing finds 8 contaminated samples among 20. Management defines the adverse state as $H = 1$ when $\theta > 0.25$. It may stop and sanitize the line (a_+) or continue production (a_-).

- a. Find the posterior distribution of θ .
- b. Compute $q = P(\theta > 0.25 \mid D)$.

Note, from R: `> pbeta(0.25, shape1 = 10, shape2 = 20, lower.tail = TRUE) → [1] 0.1663049`

- c. With $C_{FP} = \$40,000$ and $C_{FN} = \$120,000$, find the threshold, choose the Bayes action, and compare expected losses.
- d. If the scientific threshold changes from 0.25 to 0.35, recompute q and determine whether the action changes. Find the critical cost ratio for this revised state.

Note, from R: `> 1 - pbeta(0.35, shape1 = 10, shape2 = 20, lower.tail = TRUE) → [1] 0.4076133`

Q2a ... posterior distribution

Step 1. Record the counts:

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Step 1. Record the counts:

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Step 2. Update both Beta parameters:

$$\theta \mid D \sim \text{Beta}(\alpha + y, \beta + n - y).$$

Step 3. Substitute:

$$\theta \mid D \sim \text{Beta}(2 + 8, 8 + 12).$$

Answer:

$$\theta \mid D \sim \text{Beta}(10, 20).$$

Its posterior mean is $10/30 = 0.3333$.

But the decision requires a tail probability rather than the mean.

Q2b ... probability of an excessive rate

The posterior mean of 0.3333 does not directly answer the question of how likely θ is to be greater than 0.25.

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Because θ is unknown, we calculate

$$q = P(\theta > 0.25 \mid D).$$

This is the probability that the true contamination rate is greater than 25%, given the observed data.

Q2b ... probability of an excessive rate

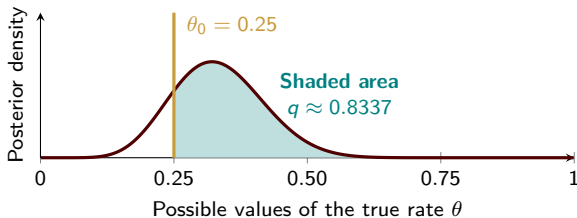


Figure: Posterior Beta(10, 20) distribution. Shaded area represents $P(\theta > 0.25 \mid D)$.

The shaded area is $q = P(\theta > 0.25 \mid D) \approx 0.8337$.

Given the model, the prior, and the test results, we assign an 83.37% posterior probability to the claim that the true contamination rate exceeds 25%.

Q2b ... probability of an excessive rate

The state concerns the underlying rate, so calculate a posterior tail probability.

Step 1. Define the relevant probability:

$$q = P(\theta > 0.25 \mid D).$$

Step 2. Use the Beta posterior CDF:

$$q = 1 - F_{\text{Beta}(10,20)}(0.25).$$

Step 3. Evaluate the CDF numerically:

$$q \approx 1 - 0.166305 = 0.833695.$$

Answer:

$$P(\theta > 0.25 \mid D) \approx 0.8337.$$

The posterior assigns an 83.37% probability to the adverse process state.

Q2c ... intervention decision

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Step 2. Compare q and τ :

$$0.8337 > 0.25.$$

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$$\tau = \frac{40,000}{40,000 + 120,000} = 0.25.$$

Step 2. Compare q and τ :

$$0.8337 > 0.25.$$

Step 3. Compare expected losses in thousands of dollars:

$$\mathcal{R}(a_+) = 40(1 - 0.833695) \approx 6.65,$$

$$\mathcal{R}(a_-) = 120(0.833695) \approx 100.04.$$

Answer:

Stop and sanitize the production line.

Q2d ... revised state threshold

Management now defines the adverse state as $\theta > 0.35$. The costs do not change.

Step 1. Recompute the state probability:

$$q_{0.35} = 1 - F_{\text{Beta}(10,20)}(0.35) \approx 0.4076.$$

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Step 2. Compare with the unchanged action threshold:

$$0.4076 > 0.25 = \tau.$$

The Bayes action remains to stop and sanitize.

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The Bayes action remains to stop and sanitize.

Step 3. Find the critical cost ratio:

$$r_{\text{crit}} = \frac{1 - q_{0.35}}{q_{0.35}} = \frac{0.5924}{0.4076} \approx 1.453.$$

Answer:

The action remains positive because the actual ratio $r = 120/40 = 3$ exceeds $r_{\text{crit}} \approx 1.453$.